

4(a)	any three from: (both processes) 1 use / require, carbon dioxide / CO₂ ; 2 use / require, water / H₂O ; 3 use / require, (sun) light energy / source of energy is light ; 4 make an organic, molecule / product ; 5 produce / release, oxygen / O₂ ; 6 involve <u>transduction</u> of energy (from one form to another) ; 7 involve electron flow ;	3
4(b)	A starch, grain / granule ; B thylakoid (membrane) / lamella / granum ;	2
4(c)(i)	ATP <u>and</u> reduced NADP (and oxygen) ;	1
4(c)(ii)	any four from: 1 electron carriers / electron transport chain / to carry electrons ; 2 to release energy to, move / pump, protons (into, thylakoid space / lumen) or to release energy to generate proton gradient ; 3 oxygen-evolving, complex / enzyme ; 4 (to catalyse) splitting / photolysis, of water ; 5 LHC / antenna complex / photosystem, <u>proteins</u> hold pigments ; 6 ATP synth(et)ase to make ATP ;	4
4(d)	triose phosphate ;	1

Question	Answer	Marks
4(e)	<i>any two plus one opposing arguments based on correct ref. to:</i> 1 percentage / amount, of sunlight, absorbed / used ; 2 absorption / action, spectrum or wavelengths / colours of light, used ; 3 respiration of organism ; 4, 5 stage/transition, where, energy lost ; ; 6, 7 (named) limiting factor(s) ; ; 8 energy storage capability or night / dark, process ; 9 quality / nutrition in / palatability, of food product ; 10 proportion edible ; 11,12 additional energy inputs ; ; 13, 14 additional resources needed ; ; 15 labour / maintenance, needed ; 16 amount of space needed ; 17 plant, hormones / coordination ; 18 evolutionary adaptation ;	3